

Data Mining Numerical Problems & Solutions

CT 72502 (Elective I) · BEI / BEX IV-I · TU, IOE · Chapter 2: Data Preprocessing

Ch 2 – Data Preprocessing – Numerical Problems & Solutions

6 distinct numerical methods · 8 PYQ instances solved · 2 extra practice problems. Every answer independently recomputed, not copied from the source notes.

Methods in this chapter

- **M1** Normalisation: min-max, z-score, decimal scaling 78 Bh
- **M2** PCA: principal components + proportion of variance 76 Ch
- **M3** Distance measures + distance matrix (Euclidean, Manhattan, Supremum) 80 Bh, 76 Ch
- **M4** Cosine similarity 80 Bh, 80 Ba, 76 Ch
- **M5** Binary similarity: Jaccard, SMC, symmetric dissimilarity 80 Ba, 76 Ch
- **M6** Schema design + OLAP operation sequence 78 Bh, 76 Ch

M1. Normalisation

Method

- **Min-max** maps v into $[new_min, new_max]$, preserves relationships exactly:

$$v' = \frac{v - min_A}{max_A - min_A} (new_max_A - new_min_A) + new_min_A$$

- **Z-score** uses mean $\bar{A}$ and std deviation σ_A . Use when min/max unknown, or outliers would squash a min-max result:

$$v' = \frac{v - \bar{A}}{\sigma_A} \quad \sigma_A = \sqrt{\frac{1}{n} \sum (v_i - \bar{A})^2}$$

- **Decimal scaling** moves the decimal point:

$$v' = \frac{v}{10^j}, \quad j = \text{smallest int such that } \max(|v'|) < 1$$

Two traps. (1) Z-score uses the **mean**, not the min. (2) For decimal scaling, j must make the max *strictly* less than 1, so $\max |v| = 1000$ needs $j = 4$, not $j = 3$.

Problem 1.1 2+2+2

78 Bh “Use the following methods to normalize the data: 200, 300, 400, 600 and 1000.” (a) min-max with $min = 0, max = 1$ (b) z-score (c) decimal scaling.

Given

$v = \{200, 300, 400, 600, 1000\}, n = 5, min_A = 200, max_A = 1000$.

(a) Min-max, $new_min = 0, new_max = 1$

$$v' = \frac{v - 200}{1000 - 200} = \frac{v - 200}{800}$$

v	v'
200	$0/800 = 0$
300	$100/800 = 0.125$
400	$200/800 = 0.25$
600	$400/800 = 0.5$
1000	$800/800 = 1$

(b) Z-score

- Mean $\bar{A} = \frac{200 + 300 + 400 + 600 + 1000}{5} = \frac{2500}{5} = 500$
- Deviations: $-300, -200, -100, +100, +500$
- $\sum (v - \bar{A})^2 = 90000 + 40000 + 10000 + 10000 + 250000 = 400000$
- $\sigma_A = \sqrt{400000/5} = \sqrt{80000} = 282.84$

v	$v' = (v - 500)/282.84$
200	$-300/282.84 = -1.0607$
300	$-200/282.84 = -0.7071$
400	$-100/282.84 = -0.3536$
600	$+100/282.84 = +0.3536$
1000	$+500/282.84 = +1.7678$

(c) Decimal scaling

- $\max |v| = 1000$.
- $j = 3$: $1000/10^3 = 1$, not < 1 . **Rejected.**
- $j = 4$: $1000/10^4 = 0.1 < 1$. **Use $j = 4$.**

v	$v' = v/10000$
200	0.02
300	0.03
400	0.04
600	0.06
1000	0.1

Final answer

- Min-max: 0, 0.125, 0.25, 0.5, 1
- Z-score: -1.0607 , -0.7071 , -0.3536 , $+0.3536$, $+1.7678$
- Decimal scaling: 0.02, 0.03, 0.04, 0.06, 0.1

Note on σ verified/added

- Above uses the **population** std deviation, $\sqrt{\sum (v - \bar{A})^2 / n}$. This is the form Han & Kamber use, so it is the safe default.
- If the examiner wants the **sample** std deviation, divide by $n - 1$: $s = \sqrt{400000/4} = 316.23$, giving -0.9487 , -0.6325 , -0.3162 , $+0.3162$, $+1.5811$.
- **State which one you used.** Either is accepted if labelled.

M2. PCA: principal components and proportion of variance**Method**

1. Start from the **covariance matrix** Σ (if given raw data: subtract the mean first, then compute Σ).
2. Solve the characteristic equation $|\Sigma - \lambda I| = 0$ for the eigenvalues λ .
3. Sort λ in **descending** order. λ_i = variance carried by component i .
4. For each λ_i solve $(\Sigma - \lambda_i I) \mathbf{v} = 0$ for the eigenvector, then normalise it to unit length.
5. Proportion of total variance explained by component i :

$$\frac{\lambda_i}{\sum_j \lambda_j}, \quad \sum_j \lambda_j = \text{trace}(\Sigma)$$

- **Check:** the eigenvalues must sum to the trace. Free and catches arithmetic slips.

Problem 2.1 4

76 Ch “Find the principal components and the proportion of the total variance explained by each when the covariance matrix of X_1, X_2, X_3 is:”

Given

$$\Sigma = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Step 1: characteristic equation

- The third row/column is decoupled, so Σ splits into a 2×2 block and a 1×1 block.
- From the 1×1 block: $(2 - \lambda) = 0 \Rightarrow \boxed{\lambda = 2}$
- From the 2×2 block:

$$(1 - \lambda)(5 - \lambda) - (-2)(-2) = 0$$

$$5 - 6\lambda + \lambda^2 - 4 = 0 \Rightarrow \lambda^2 - 6\lambda + 1 = 0$$

$$\lambda = \frac{6 \pm \sqrt{36 - 4}}{2} = \frac{6 \pm \sqrt{32}}{2} = 3 \pm 2\sqrt{2}$$

Step 2: eigenvalues, sorted

- $\lambda_1 = 3 + 2\sqrt{2} = \mathbf{5.8284}$
- $\lambda_2 = \mathbf{2}$
- $\lambda_3 = 3 - 2\sqrt{2} = \mathbf{0.1716}$
- **Check:** $5.8284 + 2 + 0.1716 = 8 = \text{trace}(\Sigma) = 1 + 5 + 2$. **Correct.**

Step 3: eigenvectors

- For $\lambda_1 = 5.8284$: row 1 gives $(1 - 5.8284)v_1 - 2v_2 = 0 \Rightarrow v_2 = -2.4142 v_1$; row 3 gives $v_3 = 0$.
 $\mathbf{e}_1 \propto (1, -2.4142, 0)$, length $\sqrt{1 + 5.8284} = 2.6131$
 $\mathbf{e}_1 = (0.3827, -0.9239, 0)$
- For $\lambda_2 = 2$: rows 1 and 2 force $v_1 = v_2 = 0$, v_3 free.
 $\mathbf{e}_2 = (0, 0, 1)$
- For $\lambda_3 = 0.1716$: row 1 gives $0.8284 v_1 = 2v_2 \Rightarrow v_2 = 0.4142 v_1$; $v_3 = 0$.
 $\mathbf{e}_3 \propto (1, 0.4142, 0)$, length 1.0824
 $\mathbf{e}_3 = (0.9239, 0.3827, 0)$
- Check:** $\mathbf{e}_1 \cdot \mathbf{e}_3 = (0.3827)(0.9239) + (-0.9239)(0.3827) = 0$
 $\rightarrow$ orthogonal, as required for a symmetric matrix.

Final answer

- PC1: $\lambda_1 = 5.8284$, $\mathbf{e}_1 = (0.3827, -0.9239, 0)$, **72.86%**
- PC2: $\lambda_2 = 2$, $\mathbf{e}_2 = (0, 0, 1)$, **25.00%**
- PC3: $\lambda_3 = 0.1716$, $\mathbf{e}_3 = (0.9239, 0.3827, 0)$, **2.14%**

Step 4: proportion of variance

PC	λ_i	$\lambda_i/8$
PC1	5.8284	72.86%
PC2	2.0000	25.00%
PC3	0.1716	2.14%
Total	8.0000	100%

Interpretation

- PC1 alone carries $\approx 73\%$ of the variance.
- PC1 + PC2 carry **97.86%** $\rightarrow$ dropping PC3 loses only 2.14%.
- So 3 dims reduce to 2 with almost no information loss.

M3. Distance measures and the distance matrix**Method**

- All three are Minkowski with a different r :

$$\text{dist}(p, q) = \left(\sum_{k=1}^n |p_k - q_k|^r \right)^{1/r}$$

- $r = 1$ **Manhattan**: $\sum |p_k - q_k|$
- $r = 2$ **Euclidean**: $\sqrt{\sum (p_k - q_k)^2}$
- $r \rightarrow \infty$ **Supremum** (L_∞ , Chebyshev): $\max_k |p_k - q_k|$
- A **distance matrix** is symmetric with a zero diagonal, so only the lower triangle needs computing.

Warm-up (Han & Kamber Example 2.19–2.20)

Smallest case that shows all three at once. Worth memorising as a check. verified/added

- $x_1 = (1, 2)$, $x_2 = (3, 5)$. Differences: 2 and 3.
- Euclidean $= \sqrt{2^2 + 3^2} = \sqrt{13} = \mathbf{3.61}$
- Manhattan $= 2 + 3 = \mathbf{5}$
- Supremum $= \max(2, 3) = \mathbf{3}$
- Ordering**: supremum $\leq$ Euclidean $\leq$ Manhattan. Always true. Use it to catch slips.

Problem 3.1 2

76 Ch “Given the following points compute the distance matrix using the Manhattan and the Supremum distance.”

Given

$P_1(6, 3)$, $P_2(2, 2)$, $P_3(3, 4)$.

Manhattan (L_1)

- $d(P_1, P_2) = |6 - 2| + |3 - 2| = 4 + 1 = 5$
- $d(P_1, P_3) = |6 - 3| + |3 - 4| = 3 + 1 = 4$
- $d(P_2, P_3) = |2 - 3| + |2 - 4| = 1 + 2 = 3$

Supremum (L_∞)

- $d(P_1, P_2) = \max(4, 1) = 4$
- $d(P_1, P_3) = \max(3, 1) = 3$
- $d(P_2, P_3) = \max(1, 2) = 2$

	P_1	P_2	P_3
P_1	0	5	4
P_2	5	0	3
P_3	4	3	0

	P_1	P_2	P_3
P_1	0	4	3
P_2	4	0	2
P_3	3	2	0

- Sanity check**: supremum $\leq$ Euclidean $\leq$ Manhattan, always. Holds here ($4 \leq 5$, $3 \leq 4$, $2 \leq 3$).

Problem 3.2 3

80 Bh “Calculate the Euclidean distance between object 1, 3 and object 1, 4.”

Given

Object	Size	Weight	Color_Code	Taste_Score
1	4	56	7	10
2	3	53	8	11
3	7	58	6	9
4	9	55	7	12

Step-by-step

- $d(O_1, O_3)$: differences $(4 - 7, 56 - 58, 7 - 6, 10 - 9) = (-3, -2, 1, 1)$

$$= \sqrt{9 + 4 + 1 + 1} = \sqrt{15} = \mathbf{3.873}$$
- $d(O_1, O_4)$: differences $(4 - 9, 56 - 55, 7 - 7, 10 - 12) = (-5, 1, 0, -2)$

$$= \sqrt{25 + 1 + 0 + 4} = \sqrt{30} = \mathbf{5.477}$$

Final answer

$d(O_1, O_3) = 3.873$, $d(O_1, O_4) = 5.477$. Object 1 is closer to object 3 than to object 4.

M4. Cosine similarity

Method

$$\cos(d_1, d_2) = \frac{d_1 \cdot d_2}{\|d_1\| \|d_2\|}, \quad d_1 \cdot d_2 = \sum_k d_{1k} d_{2k}, \quad \|d\| = \sqrt{\sum_k d_k^2}$$

- Measures **angle only**, ignores magnitude → used for documents.
- Always lay out 3 lines: dot product, then each length, then divide.

Typo in your class notes verified/added : DBK's Lecture-2 cosine example prints $\|d_2\| = (6)^{0.5} = 2.245$. In fact $\sqrt{6} = \mathbf{2.449}$. His final answer 0.3150 is correct and only follows from $2.449 (5 / (6.481 \times 2.449) = 0.315)$, so 2.245 is a slip. Copy the method, not that number.

Problem 4.1 3

80 Bh “Find the cosine similarity between Object-2 and 4.” Same table as Problem 3.2.

Given

$O_2 = (3, 53, 8, 11)$, $O_4 = (9, 55, 7, 12)$.

Step-by-step

- Dot product: $3(9) + 53(55) + 8(7) + 11(12) = 27 + 2915 + 56 + 132 = 3130$
- $\|O_2\| = \sqrt{9 + 2809 + 64 + 121} = \sqrt{3003} = 54.800$
- $\|O_4\| = \sqrt{81 + 3025 + 49 + 144} = \sqrt{3299} = 57.437$
- $\cos = \frac{3130}{54.800 \times 57.437} = \frac{3130}{3147.5}$

Final answer

$\cos(O_2, O_4) = \mathbf{0.9944}$.

- Nearly 1 → the two objects point almost the same direction.
- **Why so high:** Weight (≈ 50) dwarfs the other attributes and dominates the dot product. **Normalise before using cosine** on mixed-scale attributes, or the answer means very little. verified/added

Problem 4.2 1

80 Ba “Find the Cosine similarity between documents $d_1 = (4, 1, 2, 0, 2, 0, 0)$ and $d_2 = (2, 1, 3, 0, 1, 1, 1)$.”

- Dot: $8 + 1 + 6 + 0 + 2 + 0 + 0 = 17$
- $\|d_1\| = \sqrt{16 + 1 + 4 + 0 + 4 + 0 + 0} = \sqrt{25} = 5$
- $\|d_2\| = \sqrt{4 + 1 + 9 + 0 + 1 + 1 + 1} = \sqrt{17} = 4.1231$
- $\cos = 17/(5 \times 4.1231) = 17/20.6155$

Final answer

0.8246

Problem 4.3 1

76 Ch “Given the following two vectors compute the Cosine similarity: $D_1 = [4 \ 0 \ 2 \ 0 \ 1]$, $D_2 = [2 \ 0 \ 0 \ 2 \ 2]$.”

- Dot: $8 + 0 + 0 + 0 + 2 = 10$
- $\|D_1\| = \sqrt{16 + 0 + 4 + 0 + 1} = \sqrt{21} = 4.5826$
- $\|D_2\| = \sqrt{4 + 0 + 0 + 4 + 4} = \sqrt{12} = 3.4641$
- $\cos = 10/(4.5826 \times 3.4641) = 10/15.8745$

Final answer

0.6300

M5. Binary similarity: Jaccard, SMC, symmetric dissimilarity

Method

- Encode each object as a 0/1 vector, then count over all attribute positions:
 - M_{11} : both 1 M_{00} : both 0
 - M_{10} : $p = 1, q = 0$ M_{01} : $p = 0, q = 1$
- **SMC** (symmetric binary, 0-0 counts): $\frac{M_{11} + M_{00}}{M_{11} + M_{00} + M_{10} + M_{01}}$
- **Jaccard** (asymmetric binary, 0-0 ignored): $\frac{M_{11}}{M_{11} + M_{10} + M_{01}}$
- **Symmetric binary dissimilarity**: $d = \frac{M_{10} + M_{01}}{M_{11} + M_{00} + M_{10} + M_{01}} = 1 - SMC$
- **Nominal (non-binary) dissimilarity**: $d = (p - m)/p$, $m = \text{matches}$, $p = \text{total attrs}$.

Problem 5.1 2

76 Ch “Given the following two binary vectors compute the Jaccard similarity and Simple Matching Coefficient. $P = [0 \ 0 \ 1 \ 1 \ 0 \ 1]$, $Q = [1 \ 1 \ 1 \ 1 \ 0 \ 1]$.”

Step-by-step: tabulate position by position

Position	1	2	3	4	5	6
P	0	0	1	1	0	1
Q	1	1	1	1	0	1
Category	M_{01}	M_{01}	M_{11}	M_{11}	M_{00}	M_{11}

- $M_{11} = 3$, $M_{01} = 2$, $M_{10} = 0$, $M_{00} = 1$ (total = 6 ✓)
- $J = \frac{3}{3 + 0 + 2} = \frac{3}{5}$
- $SMC = \frac{3 + 1}{6} = \frac{4}{6}$

Final answer

$J = 0.6$, $SMC = 0.667$. SMC is higher because it credits the one 0-0 match that Jaccard discards.

Problem 5.2 2+2+2

80 Ba (i) Jaccard(Ram, Laxmi) for asymmetric binary attributes (ii) symmetric-binary dissimilarity $d(\text{Laxmi, Shyam})$ (iii) SMC(Ram, Shyam).

Given

Name	Gender	Eyecolor	Haircolor	Test-1	Test-2	Fever	Cough
Ram	M	Black	Gray	P	N	P	N
Laxmi	F	Blue	Black	P	P	N	N
Shyam	M	Blue	Gray	N	P	N	P

Step 0: split the attributes (the whole question turns on this)

- **Asymmetric binary** = Test-1, Test-2, Fever, Cough. Encode $P = 1, N = 0$. A shared "N" is uninformative, which is exactly what "asymmetric" means.
- **Symmetric** = Gender, Eyecolor, Haircolor. Both/all states are equally meaningful.
- Encoded: Ram = (1, 0, 1, 0), Laxmi = (1, 1, 0, 0), Shyam = (0, 1, 0, 1).

(i) Jaccard(Ram, Laxmi)

	T1	T2	Fev	Cou
Ram	1	0	1	0
Laxmi	1	1	0	0
Cat.	M_{11}	M_{01}	M_{10}	M_{00}

$$- M_{11} = 1, M_{01} = 1, M_{10} = 1, M_{00} = 1$$

$$- J = \frac{1}{1 + 1 + 1} = \frac{1}{3}$$

Answer

$$J = 0.333$$

(iii) SMC(Ram, Shyam)

	T1	T2	Fev	Cou
Ram	1	0	1	0
Shyam	0	1	0	1
Cat.	M_{10}	M_{01}	M_{10}	M_{01}

$$- M_{11} = 0, M_{00} = 0, M_{10} = 2, M_{01} = 2$$

$$- SMC = \frac{0 + 0}{4} = 0$$

Answer

SMC = 0 (they disagree on every test).

(ii) $d(\text{Laxmi, Shyam})$, symmetric binary attributes

- Across these 3 people each of Gender / Eyecolor / Haircolor takes exactly **2 states**, and neither state is an "absence" → all 3 are **symmetric binary**.
- Encode: Gender M=1 · Eyecolor Black=1 · Haircolor Gray=1.

	Gender	Eye	Hair
Laxmi (F, Blue, Black)	0	0	0
Shyam (M, Blue, Gray)	1	0	1
Category	s	t	s

$$- q = 0, r = 0, s = 2, t = 1 \quad (\text{total } p = 3 \checkmark)$$

- Symmetric binary dissimilarity keeps t :

$$d = \frac{r + s}{q + r + s + t} = \frac{0 + 2}{3} = \frac{2}{3}$$

Answer

$$d = 0.667$$

Cross-check: for binary attrs the symmetric formula reduces to the nominal mismatch ratio $(p - m)/p = (3 - 1)/3 = 2/3$. Same answer by both routes, so the encoding choice is safe.

verified/added

M6. Schema design + OLAP operation sequence**Method**

1. Put the **measures** in a central **fact table**, plus one foreign key per dimension.
2. Draw one **dimension table** per dimension around it.
 - **Star** → leave dimension tables flat (de-normalised).
 - **Snowflake** → normalise the hierarchies into sub-tables (location → city → province).
3. For the OLAP sequence, work from the base cuboid and ask, per dimension:
 - Need a coarser level? → **roll-up**
 - Need a finer level? → **drill-down**
 - Restricting 1 dimension? → **slice**. Restricting ≥ 2 ? → **dice**
 - Dimension not wanted at all? → **roll-up to all** (dimension reduction)
4. n dimensions → 2^n cuboids in the lattice.

Problem 6.1 3+3

78 Bh DW with dimensions *date*, *spectator*, *location*, *game* and measures *count*, *charge*. (a) Draw a star schema. (b) From the base cuboid, which OLAP operations list the total charge paid by *student* spectators at *Dashrath Stadium* in 2021?

(a) Star schema

- **Fact table** Sales: *date_key*, *spectator_key*, *location_key*, *game_key*, *count*, *charge*.
- **Dimension tables** (flat, one hop from the fact table):
 - *date*(*date_key*, *day*, *month*, *quarter*, *year*)
 - *spectator*(*spectator_key*, *name*, *status*, *phone*, *address*) where *status* ∈ {*student*, *adult*, *senior*}
 - *location*(*location_key*, *name*, *street*, *city*, *province*, *country*)
 - *game*(*game_key*, *name*, *description*, *sport_type*)
- 4 dimensions → $2^4 = 16$ cuboids in the lattice.

(b) OLAP operation sequence

1. Roll-up on *date* from *date_key* to *year*.
 2. Roll-up on *spectator* from *spectator_key* to *status*.
 3. Roll-up on *location* from *location_key* to *location_name*.
 4. Roll-up on *game* to all (the question does not split by game).
 5. Dice with *year* = 2021 and *status* = *student* and *location_name* = ``Dashrath Stadium''.
 6. Read off the *charge* measure (summed).
- **Dice, not slice** → 3 dimensions are being restricted at once.

Problem 6.2 3+3

76 Ch DW with dimensions *time*, *location*, *supplier*, *brand*, *product* and measures *count*, *price*. (a) Draw a snowflake schema. (b) From the base cuboid, which OLAP operations list total *count* for a certain *brand*, for each *state*, per year? *location* has *country/state/city*; *time* has *year/month/day*.

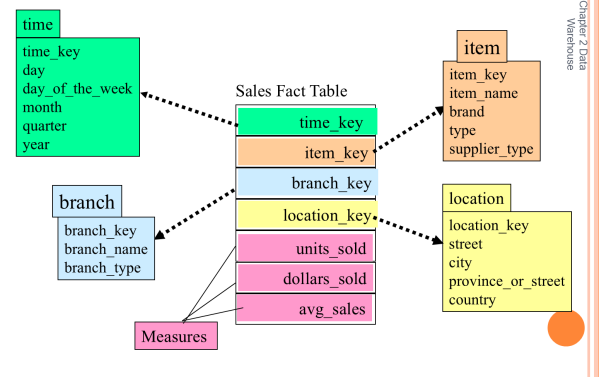
(a) Snowflake schema

- **Fact table** Sales: *time_key*, *location_key*, *supplier_key*, *brand_key*, *product_key*, *count*, *price*.
- **Normalised hierarchies** (this is what makes it a snowflake, not a star):
 - *time*(*time_key*, *day*, *month_key*) → *month*(*month_key*, *month*, *year*)
 - *location*(*location_key*, *city_key*) → *city*(*city_key*, *city*, *state_key*) → *state*(*state_key*, *state*, *country*)
 - *product*(*product_key*, *name*, *brand_key*) → *brand*(*brand_key*, *brand_name*)
 - *supplier*(*supplier_key*, *supplier_name*, *supplier_type*)
- 5 dimensions → $2^5 = 32$ cuboids.

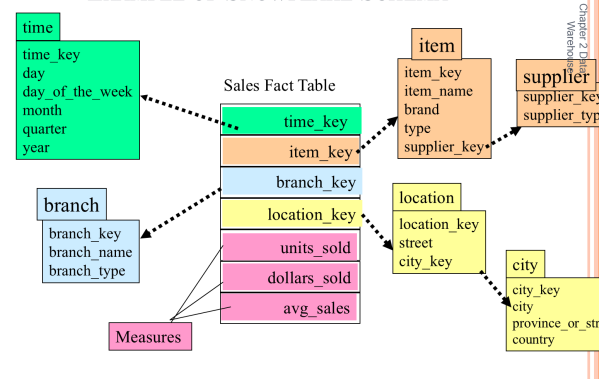
(b) OLAP operation sequence

1. Roll-up on *time* from *day* to *year*.
2. Roll-up on *location* from *city* to *state*.
3. Roll-up on *supplier* to all (not wanted).
4. Roll-up on *product* to *brand* level.
5. Slice on *brand* = the given brand (1 dimension restricted).
6. Result: cuboid [*year*, *state*] holding total count.

EXAMPLE OF STAR SCHEMA



EXAMPLE OF SNOWFLAKE SCHEMA



Extra practice (patterns not yet fully covered)

Practice A — binning for noise smoothing

Binning is examined in prose 72 Ka, 76 Ash but never as a numerical. **This is DBK's own worked example** (Lecture-1, "Binning Methods for Data Smoothing"), so it is the likeliest form any future numerical would take. Answers below independently recomputed and they match his slide exactly.

Problem

Sorted price data: 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34. Partition into **equal-depth** bins of depth 4 and smooth by (i) bin means (ii) bin boundaries.

Solution

- 12 values, depth 4 $\rightarrow$ 3 bins.
 - Bin 1: 4, 8, 9, 15 Bin 2: 21, 21, 24, 25 Bin 3: 26, 28, 29, 34
- (i) **Smoothing by bin means**
 - Bin 1 mean = $36/4 = 9 \rightarrow$ **9, 9, 9, 9**
 - Bin 2 mean = $91/4 = 22.75 \approx 23 \rightarrow$ **23, 23, 23, 23**
 - Bin 3 mean = $117/4 = 29.25 \approx 29 \rightarrow$ **29, 29, 29, 29**
- (ii) **Smoothing by bin boundaries** (each value $\rightarrow$ nearer of min/max)
 - Bin 1 (min 4, max 15): 4, 8, 9, 15 $\rightarrow$ **4, 4, 4, 15** (8 and 9 are nearer 4)
 - Bin 2 (min 21, max 25): 21, 21, 24, 25 $\rightarrow$ **21, 21, 25, 25**
 - Bin 3 (min 26, max 34): 26, 28, 29, 34 $\rightarrow$ **26, 26, 26, 34**
- **Equal-width** for contrast: range = $34 - 4 = 30$, $N = 3 \rightarrow W = (B - A)/N = 10$. Intervals $[4, 14)$, $[14, 24)$, $[24, 34]$ $\rightarrow$ counts 3, 3, 6. **Unequal counts: that is the difference from equal-depth.**

Practice B — min-max into a range other than [0,1]

Every PYQ so far uses $[0, 1]$. The figures below are Han & Kamber 3rd ed Example 3.4, so they are the canonical ones. verified/added

Problem

Income values 12,000 to 98,000. Normalise **73,600** to the range $[0, 1]$, then to $[-1, 1]$.

Solution

- To $[0, 1]$: $v' = \frac{73600 - 12000}{98000 - 12000}(1 - 0) + 0 = \frac{61600}{86000} = \mathbf{0.716}$
- To $[-1, 1]$: $v' = \frac{61600}{86000}(1 - (-1)) + (-1) = 0.716(2) - 1 = \mathbf{0.432}$
- **Check:** the value sits 71.6% of the way up the range, so in $[-1, 1]$ it must land slightly above the midpoint 0. It does.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity–relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	<i>k</i> -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
→	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /17** = topic asked in 7 or more of the 17 papers · **HOT /17** = 4–6 papers · **PIN /17** = 1–3 papers but worth ≥ 8 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **verified/added** marks anything researched and added that was not in the source notes.